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Impact of artificial intelligence-enabled service attributes on customer satisfaction and loyalty in chain hotels: Evidence from coastal tourism destinations in western Thailand

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ABSTRACT

This study investigates how artificial intelligence (AI)-enabled service attributes affect customer satisfaction and loyalty in chain hotels located in coastal tourism destinations in Western Thailand. We integrate the technology acceptance model, SERVQUAL, the stimulus-organism-response model, and the humanness-value-loyalty model to examine how factors like reliability, perceived intelligence, service efficiency, aesthetics, and enjoyment affect satisfaction with AI-based customer services and customer loyalty. Data from 186 hotel guests were analyzed using partial least squares structural equation modeling. The results reveal that service efficiency ($\beta = 0.251$, $p < 0.01$) and enjoyment ($\beta = 0.526$, $p < 0.001$) significantly influence satisfaction with AI-based services. Additionally, satisfaction strongly affected customer loyalty ($\beta = 0.833$, $p < 0.001$); however, reliability, perceived intelligence, and aesthetics did not significantly influence satisfaction. The model demonstrates strong explanatory power, with R^2 values of 0.765 for satisfaction and 0.694 for loyalty. Mediation analysis shows that satisfaction significantly mediates the relationships between enjoyment and loyalty and between service efficiency and loyalty. These findings provide valuable insights for hotel managers implementing AI technologies, emphasizing the importance of creating efficient and enjoyable AI interactions to enhance customer satisfaction and foster long-term loyalty. This study contributes to the growing literature on AI in hospitality by comprehensively elucidating how AI-enabled service attributes influence customer outcomes in a specific market context.

1. Introduction

Artificial intelligence (AI) has transformed the hospitality industry, reshaping customer service interactions and potentially influencing guest loyalty (Belanche et al., 2021; Hsu & Lin, 2023). As chain hotels in coastal tourism destinations increasingly adopt AI-enabled services, understanding their impact on customer loyalty is crucial to maintaining competitiveness and ensuring long-term success (Meeprom et al., 2024). Integrating AI technologies into hotel operations has enhanced operational efficiency, personalized guest experiences, and improved service quality (Chen et al., 2023; Liu et al., 2024). These advancements can significantly influence customer satisfaction and loyalty, key drivers of profitability in the hospitality sector (Rahman et al., 2021). AI adoption in the hospitality industry has been extensively researched, with studies

exploring various aspects of AI implementation, including its effects on service quality, customer satisfaction, and behavioral intentions (Chiang et al., 2021; Pizzi et al., 2023). For example, Hsu and Lin (2023) investigated how AI enhances personalization and improves response times in hotel services, and Al-Hyari et al. (2023) examined how AI-powered 24/7 customer support affects guest satisfaction.

Hospitality research has employed several theoretical frameworks to understand the relationship between AI-enabled services and customer loyalty. The technology acceptance model (TAM) has been widely used to explain user adoption of new technologies, including AI-powered hotel services (Belanche et al., 2021). Furthermore, the SERVQUAL model assesses service quality across five dimensions: tangibles, reliability, responsiveness, assurance, and empathy; it has been adapted to evaluate AI-enabled service quality in hospitality settings (Choi et al.,

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2024). Additional research has applied the stimulus-organism-response (S-O-R) framework to examine how AI-enabled service features (stimuli) influence customer perceptions and behaviors (Hsu & Lin, 2023). Recent studies have also introduced new theoretical perspectives, including the humanness-value-loyalty (HVL) model proposed by Chen et al. (2021), which explores how perceived humanness in AI interactions affects customer value perceptions and loyalty. Furthermore, Rajaobelina et al. (2021) investigated how reliability, perceived intelligence, service efficiency, aesthetics, and enjoyment shape customer satisfaction and loyalty regarding AI-based services.

Despite the growing research on AI in the hospitality industry, understanding how AI-enabled services influence customer loyalty across specific geographic and market segments, particularly in emerging markets and coastal tourism destinations remains underexplored (Hansasooksin et al., 2024; Hansasooksin & Tontisirin, 2021). Previous studies provided valuable insights into various aspects of AI adoption in hospitality; however, limited research has integrated multiple theoretical perspectives to comprehensively assess the impact of AI-enabled services on customer loyalty in chain hotels situated in coastal tourism destinations.

By focusing on coastal tourism destinations in emerging markets such as Thailand, this study addresses a critical research gap regarding AI implementation in hospitality. These destinations face unique operational challenges that set them apart from urban or inland hotels. Coastal tourism areas experience extreme fluctuations in seasonal demand that necessitate adaptive service delivery (Goffi et al., 2021; Hansasooksin & Tontisirin, 2021). They also cater to a diverse international clientele with varying technological preferences and language needs (Liu et al., 2024). These locations also often struggle as a result of the limited availability of skilled labor (Wong et al., 2024), while weather-dependent operations impact service consistency (Hansasooksin et al., 2024).

AI-enabled services can address these challenges by offering automated multilingual support, dynamic resource allocation, and consistent service delivery, regardless of staffing levels or seasonal constraints (Al-Hyari et al., 2023; Chen et al., 2023). Previous research on AI in the context of hospitality has primarily concentrated on urban hotels in developed markets (Belanche et al., 2021; Hsu & Lin, 2023), while the effectiveness of AI in coastal tourism contexts where operational conditions and customer expectations may differ significantly remains underexplored (Liu et al., 2024; Meeprom et al., 2024). This lack of understanding regarding the performance of AI technologies in these unique environments underscores the necessity for targeted research examining AI implementation in coastal tourism settings.

This study addresses this research gap by focusing on chain hotels in coastal tourism destinations of Western Thailand—an area that has received limited attention. The research integrates key theoretical frameworks such as the Technology Acceptance Model (TAM), SERVQUAL, the Stimulus-Organism-Response (S-O-R) model, and the Hierarchical Value Map (HVL) to provide a novel approach to understanding the complex relationships between AI-enabled services and customer loyalty within a specific geographic and market context.

The rapid growth of AI adoption in the hospitality industry, along with the unique characteristics of coastal tourism destinations, underscores the significance of this study. As competition intensifies and customer expectations evolve, hotel managers and policymakers must comprehend how AI-enabled services influence customer loyalty (Al-Hyari et al., 2023; Wong et al., 2024). This research is particularly relevant in emerging markets like Thailand, where AI adoption in hospitality is still in its early stages, and its impact on customer loyalty has yet to be fully explored (Goffi et al., 2021; Liu et al., 2024). We focus on chain hotels in coastal tourism areas because of their economic importance and examine how AI can efficiently address unique challenges such as seasonal demand fluctuations while serving a diverse international clientele.

By examining this specific context, the study aims to provide

theoretically robust and practically applicable insights for an essential segment of the hospitality industry. This study primarily investigates how AI-enabled services impact customer loyalty in chain hotels located in coastal tourism destinations in Western Thailand. Specifically, we.

- (1) Examine the relationships between the key dimensions of AI-enabled services and customer satisfaction;
- (2) Assess the mediating role of customer satisfaction in the relationship between AI-enabled services and customer loyalty;
- (3) Develop and validate an integrated model that explains how AI-enabled services affect customer loyalty in the context of chain hotels in coastal tourism destinations.

This study makes several contributions to the existing literature. First, this research integrates multiple theoretical perspectives to comprehensively understand the factors influencing customer loyalty regarding AI-enabled hotel services. Second, we offer insights into the effectiveness of AI-enabled services within a specific geographic and market context, addressing the scarcity of research in emerging markets and coastal tourism destinations. Finally, this study develops and validates an integrated model to guide future research and inform managerial decision-making regarding AI implementation in hospitality. The research employs a quantitative approach, utilizing surveys to achieve these objectives. Data were collected from customers of chain hotels in Western Thailand's coastal tourism destinations, and a large-scale survey of hotel guests was conducted to test the proposed theoretical model. The findings offer valuable guidance for hotel managers and policymakers to leverage AI technologies to improve customer experiences and encourage long-term loyalty in this highly competitive industry.

The remainder of this paper is structured to systematically examine the relationship between AI-enabled services and customer loyalty in the hospitality industry. Section 2 provides a comprehensive literature review, synthesizing this domain's relevant theories and empirical findings. In Section 3, we develop an integrated theoretical framework and formulate our research hypotheses based on the identified gaps in current knowledge. The methodology section (Section 4) explains our research design, data collection procedures, and analytical techniques. Section 5 then presents the quantitative and qualitative findings in the results section, followed by a thorough discussion and exploration of the theoretical implications and practical applications. Section 6 summarizes the key findings, while Section 7 acknowledges the study's limitations and suggests promising future research directions.

2. Theoretical underpinning and hypothesis development

This study combines four complementary theoretical frameworks to comprehensively examine AI-enabled services in the hospitality industry. Each model highlights how the attributes of AI services influence customer satisfaction and loyalty. The Technology Acceptance Model (TAM) explains user adoption and acceptance of AI technologies by assessing perceived usefulness and ease of use (Belanche et al., 2021). SERVQUAL evaluates service quality dimensions adapted for AI contexts (Choi et al., 2024). The Stimulus-Organism-Response (S-O-R) framework illustrates how AI service attributes (stimuli) affect customer psychological states and subsequent behaviors (Hsu & Lin, 2023). Lastly, the Humanness Value Level (HVL) model explores how perceived human-like qualities in AI interactions shape customer value perceptions and loyalty (Chen et al., 2021). These models combine to create a holistic framework for understanding the technical (TAM), service quality (SERVQUAL), psychological (S-O-R), and human-AI interaction (HVL) elements that impact customer satisfaction and loyalty in AI-enabled hospitality services. This integrated theoretical approach allows us to explore the functional and experiential dimensions of AI service encounters in coastal tourism settings.

2.1. Technology acceptance model (TAM)

The TAM offers a valuable framework for understanding how customers adopt and accept new technologies, including AI-enabled services in the hospitality industry. Numerous studies have employed TAM to analyze customer acceptance of AI and robotics in hotel settings. For example, [Belanche et al. \(2021\)](#) found that customers' perceptions of competence and warmth in service robots positively impacted their expectations of service value and intentions to remain loyal. Similarly, [Hsu and Lin \(2023\)](#) demonstrated that engaging with AI voice assistants enhanced consumers' enthusiasm for the technology, affecting their intimacy, commitment, and loyalty toward services. Key constructs of TAM, such as perceived usefulness and ease of use of AI technologies, have significantly influenced customer satisfaction and behavioral intentions ([Chen et al., 2021](#); [Chiang et al., 2021](#)). Regarding perceived intelligence, [Hasan et al. \(2021\)](#) revealed that consumers' trust and the novelty value of AI voice assistants positively influenced brand loyalty, while perceived risk had a negative impact. On the subject of service efficiency (SER), [Zhan et al. \(2021\)](#) found that AI-enabled personalization and operational enhancements in retail pharmacies improved customer experience and loyalty. In the hotel context, [Fernández-Rovira et al. \(2021\)](#) argued that AI and big data analytics enable more efficient and personalized services, potentially increasing customer satisfaction. At the same time, Lee et al. (2021) warned that while AI can enhance service efficiency, the perceived warmth and human element remain crucial factors in how customers evaluate services. Overall, the extant literature indicates that perceived intelligence and SER (as facilitated by AI technologies) are essential in shaping customer acceptance, satisfaction, and loyalty within the hospitality sector, aligning with the fundamental principles of TAM.

2.2. SERVQUAL model

The SERVQUAL model has been widely used to assess service quality across various industries, particularly hospitality ([Parasuraman et al., 1988](#)). Recent research has examined adapting this model to evaluate AI-enabled services, especially in hotels ([Belanche et al., 2021](#); [Hsu & Lin, 2023](#)). The key dimensions influencing customer acceptance and satisfaction with AI-enabled hotel services include reliability (REL) and SER. REL refers to the consistency and dependability of AI systems in accurately delivering promised services ([Chiang et al., 2021](#)). For example, AI-powered chatbots must provide accurate information and fulfill guest requests to build trust ([Pizzi et al., 2023](#)). SER pertains to the speed and ease with which AI systems can complete tasks and assist guests ([Wu et al., 2021](#)). Technologies like voice assistants can enhance efficiency by quickly responding to queries and automating processes ([Hernandez-Ortega & Ferreira, 2021](#)). Studies indicate that satisfaction and loyalty increase when hotel guests perceive AI services as both reliable and efficient ([Belanche et al., 2021](#); [Hsu & Lin, 2023](#)); however, challenges persist in balancing efficiency with personalization and maintaining the "human touch" in hospitality experiences ([Plattfaut & Koch, 2021](#)). Some research suggests that while AI excels in handling routine tasks, human staff may still be preferred for complex or emotionally nuanced interactions ([Chen et al., 2021](#)). As AI capabilities continue to advance, hotels must carefully consider integrating these technologies to enhance service quality without compromising the emotional connections and personalized experiences fundamental to hospitality ([Mouchabac et al., 2021](#)). Additional research is necessary to refine the SERVQUAL model for AI-enabled services and to understand how different AI implementations affect guest perceptions of reliability and efficiency in the changing landscape of hotel services.

2.3. Stimulus-organism-response (S-O-R) framework

The S-O-R framework offers a valuable perspective for understanding how AI service attributes influence customer psychological states and

behaviors in the hospitality industry. This framework posits that environmental stimuli (S) affect customers' internal states (O), which subsequently drive their behavioral responses (R) ([Belanche et al., 2021](#); [Choi et al., 2024](#)). Two key stimuli—*aesthetics* (AES) and *enjoyment* (ENJ)—shape how customers perceive and experience AI-enabled hotel services. These AI service attributes are external cues that trigger internal cognitive and emotional processes ([Hsu & Lin, 2023](#)). For example, a visually appealing and enjoyable AI interface can elicit positive emotions and enhance perceived usefulness, representing the organism component of the S-O-R model ([Chen et al., 2021](#)). These internal states then influence behavioral outcomes, such as customer loyalty (LOY) and intentions to continue using the service ([Hernandez-Ortega & Ferreira, 2021](#); [Rajaobelina et al., 2021](#)). The S-O-R framework has been utilized to explore how AI-driven service interactions impact customer satisfaction and loyalty across various hospitality contexts ([Al-Hyari et al., 2023](#); [Belanche et al., 2021](#)). Research has shown that AI attributes like perceived intelligence and anthropomorphism can positively influence customer trust, satisfaction, and loyalty by mediating cognitive and emotional responses ([Choi et al., 2024](#); [Hsu & Lin, 2023](#)). Moreover, the S-O-R framework clarifies how AI-enabled personalization and service recovery can shape customer engagement and loyalty by influencing customers' emotional and cognitive reactions ([Rajaobelina et al., 2021](#)). In summary, the S-O-R model provides a comprehensive approach to understanding the intricate relationships between AI service stimuli, customer internal states, and behavioral responses in the evolving landscape of AI-enabled hospitality services.

2.4. Humanness-value-loyalty (HVL) model

The HVL model offers a framework for understanding how customers perceive and respond to AI-enabled services, particularly within the hospitality industry. This model examines the relationships among perceived intelligence, aesthetics, customer value perceptions, and loyalty intentions. [Belanche et al. \(2021\)](#) proposed and empirically validated the HVL model, analyzing how customers' perceptions of physical likeness, competence, and warmth regarding robots can influence service value expectations and loyalty intentions. They found that human likeness positively influenced several dimensions of service value expectations, perceived competence predominantly impacted utilitarian expectations, and perceived warmth affected relational expectations. In the context of AI-enabled hotel services, the HVL model can be used to understand how the perceived intelligence and aesthetics of AI services can affect customer acceptance and satisfaction. [Hasan et al. \(2021\)](#) showed that consumer trust, interaction, and novelty positively influenced brand loyalty for AI-supported devices, while perceived risk negatively affected loyalty. Similarly, [Hernandez-Ortega and Ferreira \(2021\)](#) found that engaging experiences with AI voice assistants enhanced consumers' passion for technology, increasing intimacy and commitment and ultimately fostering service loyalty. [Rajaobelina et al. \(2021\)](#) underscored the significance of customer value perceptions in AI-enabled services. They discovered that feelings of creepiness could diminish loyalty and have an indirect negative effect on trust and negative emotions. This finding underscores the importance of carefully considering AI aesthetics and perceived intelligence to avoid adverse customer reactions. Finally, [Chiang et al. \(2021\)](#) highlighted the role of collective intelligence and dissonance in social shopping behaviors, which can be applied to AI-enabled hotel services to enhance customer engagement and loyalty. In summary, the HVL model provides a valuable framework for understanding how AI services' perceived intelligence and aesthetics influence customer value perceptions and loyalty intentions in the hospitality industry. Hotels can consider these elements to design and implement AI-enabled services that improve customer acceptance, satisfaction, and loyalty.

2.5. Role of AI chatbots in enhancing customer loyalty

Companies increasingly adopt AI chatbots to enhance customer service and build loyalty. These conversational AI agents use natural language processing to interact with customers in a human-like manner (Hsu & Lin, 2023). Research indicates that when implemented effectively, AI chatbots can positively affect customer satisfaction, engagement, and loyalty. Chatbots can improve customer service by offering 24-h availability, rapid response times, and personalized interactions (Chiang et al., 2021). The convenience and efficiency of chatbot interactions often lead to higher customer satisfaction than traditional service channels (Chen et al., 2022); however, the perceived quality of the chatbot service is crucial—customers expect chatbots to be competent, responsive, and capable of understanding their needs (Hsu & Lin, 2023). The human-like qualities of AI chatbots also play a significant role. Research shows that chatbots with higher levels of perceived anthropomorphism, warmth, and social presence can foster stronger emotional connections with customers (Belanche et al., 2021). This connection can increase trust, engagement, and intentions to remain loyal, providing new opportunities for customer relationship management; however, human-like chatbots can also instill feelings of eeriness or "creepiness," which may harm customer relationships (Rajabbelina et al., 2021).

Studies have explored how chatbot interactions influence various dimensions of customer loyalty. For example, cognitive loyalty can significantly increase when chatbots provide helpful information and efficiently resolve customer issues (Chiang et al., 2021). This practical benefit reassures customers about the value of AI chatbots. Furthermore, affective loyalty may develop when customers have positive emotional experiences with personable chatbots (Belanche et al., 2021). Conative loyalty, or the intention to repurchase, increases when chatbots provide high-quality service (Hsu & Lin, 2023); however, research also highlights challenges in building customer loyalty through AI chatbots. Privacy concerns, a lack of emotional intelligence, and an inability to handle complex queries can negatively affect the customer experience (Rajabbelina et al., 2021). Additional studies suggest that chatbots are better suited for transactional interactions than building relationships (Chen et al., 2022). Overall, the literature indicates that AI chatbots have significant potential to enhance customer loyalty when designed and implemented strategically. The critical factors for success include perceived intelligence, social presence, service quality, and the ability to create positive emotional experiences. Nevertheless, companies must also be aware of the potential downsides that could damage customer relationships. As AI technology advances, further research is needed to understand how such technology can most effectively foster long-term customer loyalty across various contexts and industries.

2.6. AI chatbots and hotel customer loyalty

Hotels increasingly adopt AI chatbots to enhance customer service and foster loyalty, and several studies have investigated the impact of AI chatbots on customer experiences and loyalty within the hospitality industry. For example, Chiang et al. (2021) found that the quality of AI-chatbot services positively influences hotel customer satisfaction and loyalty. They emphasized that chatbots could offer personalized recommendations and provide 24/7 support, thereby improving the overall customer experience. Similarly, Meeprom et al. (2024) reported that AI-driven customer engagement positively affected loyalty, although interactions with human staff remain essential. They suggested that a hybrid model combining AI and human service could be the most effective approach.

Additionally, perceptions of human-like qualities and intelligence in AI chatbots can influence customer reactions. For example, Belanche et al. (2021) developed an HVL model that illustrates how perceptions of a chatbot's human likeness, competence, and warmth affect customer expectations of service value and loyalty intentions. Notably, they found

that warmth had a more significant impact on customers with lower social interaction needs. Hernandez-Ortega and Ferreira (2021) also underscored the importance of emotional connections, revealing that consumers can develop feelings of "love" toward AI voice assistants, which can drive loyalty. Conversely, some research highlights the potential drawbacks of AI chatbots. For instance, Rajabbelina et al. (2021) discovered that perceptions of "creepiness" toward chatbots could decrease trust and loyalty, particularly when privacy concerns are heightened. Hasan et al. (2021) reported that the perceived risks associated with AI voice assistants negatively affect brand loyalty; thus, hotels must carefully manage customer perceptions and privacy concerns when implementing AI chatbots.

The role of AI chatbots may also differ based on customer segments and hotel types. For example, Al-Araj et al. (2022) found that AI positively influenced loyalty more in luxury hotels than in mid-range properties. Similarly, Li et al. (2023) noted that customer innovativeness and involvement moderate the effects of AI chatbots on loyalty, indicating that hotels may need tailored AI strategies for different customer groups. The literature suggests that AI chatbots in hotels can effectively enhance customer loyalty. Key success factors include perceived intelligence, warmth, and human-like qualities. At the same time, addressing privacy concerns and "creepiness" remains crucial. A hybrid model that combines AI and human service may present the optimal approach; however, more research is necessary to understand the long-term impacts and the best methods for integrating AI chatbots with other hotel services. As AI technology advances, its role in building customer relationships and loyalty in the hospitality industry will likely grow.

2.7. Hypothesis development

Reliability is a crucial service quality dimension that significantly shapes customer perceptions and experiences with AI-powered technologies, such as chatbots and virtual assistants. Numerous studies have emphasized the importance of reliability in these AI-based services and their impact on customer satisfaction. For example, Hsu and Lin (2023) found that the core service quality of AI chatbots (with reliability as a critical component) significantly influenced user satisfaction. Their research on customer service chatbots revealed that consistent and reliable performance is essential for building user trust and satisfaction. Similarly, Chen et al. (2023) demonstrated that system stability, a vital aspect of reliability, positively affects customer satisfaction and loyalty in AI chatbots, suggesting that a stable and dependable AI system is crucial for fostering positive customer attitudes.

In the hospitality industry, Meeprom and Suttikun (2024) found that the performance value of AI-based services significantly affects customer engagement, influencing customer loyalty. They focused on restaurant services, highlighting the critical role of reliable AI technologies in shaping customer perceptions and behaviors. Additionally, Oliinyk et al. (2022) discussed how implementing AI in hotels can improve operational efficiency and customer satisfaction, emphasizing the importance of reliable AI systems in enhancing the overall guest experience. Rajabbelina et al. (2021) supported the relationship between reliability and customer satisfaction. They found that chatbot usability, including reliability, reduces discomfort and indirectly impacts customer loyalty through trust. This finding indicates that reliable AI interactions can alleviate negative emotions and enhance the customer experience. Belanche et al. (2021) also emphasized the significance of perceived competence (closely tied to reliability) in shaping utilitarian expectations and perceptions of service value when interacting with frontline robots.

Studies examining the consequences of unreliable systems underscore the critical need for dependability in AI-based services. For instance, Fernández-Rovira et al. (2021) highlighted that employing AI and big data in customer interactions raises ethical concerns, stressing the necessity for reliable and trustworthy AI systems to maintain

customer confidence. Similarly, [Mouassa et al. \(2021\)](#) emphasized the importance of reliable AI-powered energy management systems in hotels to ensure guest satisfaction and operational efficiency. Finally, [Liu et al. \(2024\)](#) found that the reliability of AI-powered services significantly influences customer trust and loyalty in the context of intelligent hotels. They emphasized the need for hospitality businesses to ensure the reliability of their AI technologies to meet customer expectations and build long-term relationships. Given these findings, we propose the following hypothesis.

H1. Reliability positively influences satisfaction with AI-based customer services.

Perceived intelligence has become crucial to customer satisfaction with AI-driven services; as AI technologies become more widespread in hospitality settings, research indicates that customers' perceptions of an AI system's intelligence significantly affect their satisfaction with service experiences. [Belanche et al. \(2021\)](#) developed an HVL model, demonstrating that the perceived competence of AI service robots positively influences customers' expectations for utilitarian value. In turn, these expectations affect satisfaction and loyalty. Similarly, [Hsu and Lin \(2023\)](#) found that the quality of conversational AI chatbots (closely related to perceived intelligence) significantly impacts customer satisfaction.

The concept of perceived intelligence includes various factors, such as an AI system's ability to understand and respond appropriately to queries, provide relevant information, and solve problems efficiently. [Chiang et al. \(2021\)](#) noted that the perceived AI capabilities of chatbots, including their ability to engage in intelligent dialogue, positively affected customer satisfaction. In the hospitality context, [Bovsh et al. \(2022\)](#) found that when AI-powered chatbots and hotel virtual assistants were perceived as intelligent and capable of offering personalized recommendations and assistance, they contributed to guest satisfaction. Additionally, [Meeprom and Suttikun \(2024\)](#) observed that AI-based services in restaurants positively affected customer satisfaction, with perceived intelligence being a key factor. Various other factors often mediate the relationship between perceived intelligence and satisfaction, and trust has been identified as a mediator; when customers perceive AI systems as intelligent, their trust in the technology increases, leading to greater satisfaction ([Hasan et al., 2021](#)). [Hernandez-Ortega and Ferreira \(2021\)](#) discovered that innovative experiences with AI voice assistants, which incorporate perceptions of intelligence, influence emotional connections (conceptualized as "love") with the technology, ultimately impacting consumers' satisfaction and loyalty.

Nonetheless, the impact of perceived intelligence on satisfaction can vary depending on the specific application and context. For example, [Choi et al. \(2024\)](#) found that while perceived intelligence generally had a positive effect, its impact was moderated by whether a human or AI agent provided the service. They suggested that high perceived intelligence might be more impactful for AI agents than human agents in some scenarios. Additionally, [Pizzi et al. \(2023\)](#) cautioned that while a chatbot's perceived intelligence positively influences customer satisfaction, perceived warmth must also be conveyed to avoid potential adverse effects, such as increased skepticism. The positive influence of perceived intelligence on satisfaction can benefit hospitality businesses significantly. Increased satisfaction from intelligent AI interactions has been linked to greater customer loyalty, positive word-of-mouth, and increased purchase intentions ([Wei et al., 2021](#); [Zhan et al., 2021](#)). This influence underscores the strategic importance of investing in and highlighting the intelligence of AI-based customer service solutions in hospitality businesses. Based on these findings, we propose the following hypothesis.

H2. Perceived intelligence positively influences satisfaction with AI-based customer services.

The extant hospitality literature has firmly established the relationship between service efficiency and customer satisfaction, and emerging

research indicates that this relationship also applies to AI-based customer services. Service efficiency refers to the speed, accuracy, and overall quality with which services are delivered to customers ([Belanche et al., 2021](#)). In AI-based services, such as chatbots and virtual assistants, efficiency is often measured by response time, understanding and addressing customer queries, and the seamless completion of tasks ([Hsu & Lin, 2023](#)). Numerous studies have found that customers' satisfaction increases when they perceive AI-powered services as efficient.

For example, [Chen et al. \(2022\)](#) studied AI chatbots in the hotel industry, revealing a positive correlation between customer satisfaction scores and the perceived efficiency of the chatbot—specifically, the speed of responses and the accuracy of provided information. Similarly, [Rajaobelina et al. \(2021\)](#) examined consumer interactions with AI chatbots for car insurance quotes, determining that the usability and efficiency of the chatbot significantly influenced customer satisfaction and loyalty intentions. In the broader context of hospitality, [Meeprom and Suttikun \(2024\)](#) analyzed both AI-based and employee-based customer services in restaurants, finding that efficiency was a key driver of satisfaction for both service types. Furthermore, the positive relationship between service efficiency and customer satisfaction was attributed to AI's ability to deliver quick, convenient, personalized assistance. [Chiang et al. \(2021\)](#) noted that AI-powered systems could rapidly process large amounts of data to provide tailored recommendations and solutions, enhancing the perceived efficiency of service interactions. Their finding aligns with [Noor et al. \(2022\)](#), who developed a scale to measure AI service quality and identified efficiency as a critical dimension influencing customer satisfaction and loyalty.

Notably, service efficiency's impact on satisfaction may be even more pronounced for AI-based services than traditional human-delivered services. For example, [Belanche et al. \(2021\)](#) proposed that customers have heightened expectations regarding the speed and accuracy of AI systems, making efficiency a critical factor in their satisfaction evaluations. [Hsu and Lin \(2023\)](#) corroborate this finding, revealing that the perceived efficiency of AI voice assistants had a more significant influence on customer satisfaction than other factors, such as anthropomorphism or emotional connection. The positive impact of service efficiency on satisfaction extends across various types of AI applications in hospitality. For instance, [Liu et al. \(2024\)](#) determined that the efficiency of AI-driven amenities and services in intelligent hotels significantly affected guest satisfaction and loyalty intentions.

Further research suggests that service efficiency directly impacts satisfaction and can influence it indirectly through other mediating factors. For example, [Wei et al. \(2022\)](#) found that efficient AI-powered services in online travel agencies enhanced customer trust, leading to higher satisfaction levels. Similarly, [Hernandez-Ortega and Ferreira \(2021\)](#) discovered that efficient interactions with intelligent voice assistants fostered emotional connections and a "consumer love" for the technology, ultimately improving satisfaction and loyalty. Based on these findings, we propose the following hypothesis.

H3. Service efficiency positively influences satisfaction with AI-based customer services.

As AI and robotic technologies become increasingly common in customer-facing roles, their physical appearance and design aesthetics significantly influence consumer perceptions and experiences. [Belanche et al. \(2021\)](#) proposed the HVL model, suggesting that frontline service robots' perceived human-like appearance positively affects service value expectations across functional, monetary, and emotional dimensions. Their experimental study indicated that the human-like appearance of robots significantly influenced both utilitarian and relational value expectations, ultimately increasing loyalty. Similarly, [Reig et al. \(2021\)](#) examined how AI assistants' embodiment and visual design affect user perceptions, finding that customers generally preferred embodied robots to virtual agents in complex problem-solving scenarios. The importance of visual aesthetics extends beyond anthropomorphism to the overall interface design. [Hsu and Lin \(2023\)](#) investigated the factors influencing

satisfaction with AI-powered chatbots. They identified aesthetic appeal as a critical component of chatbot service quality that positively affects user satisfaction and loyalty, arguing that an aesthetically pleasing and visually engaging chatbot interface could enhance the overall user experience and foster positive perceptions of the AI assistant.

Regarding voice-based AI, [Hernandez-Ortega and Ferreira \(2021\)](#) found that positive experiences with voice assistants (including aesthetic qualities) nurtured feelings of intimacy toward AI assistants, which ultimately promoted service loyalty. Several studies have highlighted the role of robot aesthetics in shaping guest experiences in the hospitality industry. For example, [Zhong and Verma \(2019\)](#) conducted experiments with hotel service robots, finding that guests favored robots with human-like features over those with machine-like designs. Similarly, [Tussyadiah and Park \(2018\)](#) found that hotel guests perceived greater social presence in humanoid service robots than in nonhumanoid designs, leading to higher satisfaction with the service interaction. These findings emphasize that AI interfaces' visual aesthetics and anthropomorphism can evoke positive social responses in customers, enhancing their service experience in hospitality settings.

The aesthetic design of digital AI interfaces, such as chatbots and virtual assistants, is also critical in the hospitality sector. [Choi et al. \(2024\)](#) examined how various aesthetic features of AI chatbots influenced hotel customer responses, revealing that the inclusion of human-like visual cues and personalized interfaces increased customer engagement and satisfaction with AI-mediated services. [Buhalis and Sinarta \(2019\)](#) discussed how visual presentation and user experience design of AI-powered hotel services could shape guest perceptions and acceptance, arguing that an aesthetically appealing and seamlessly integrated AI interface could foster guest satisfaction and loyalty. While [Ivanov and Webster \(2019\)](#) cautioned against prioritizing aesthetics over service quality, thoughtful implementation of aesthetic design can be a powerful tool for enhancing AI-based services' appeal and user experience. [Wirtz et al. \(2018\)](#) noted that the "servicescape," which includes visual and aesthetic elements, is crucial in shaping customers' holistic service experiences with AI technologies in hospitality settings. Based on these findings, we propose the following hypothesis.

H4. Aesthetics positively influence satisfaction with AI-based customer services.

Researchers have increasingly focused on the relationship between enjoyment and satisfaction with AI-based customer services in the hospitality industry. As AI technologies become more common in customer-facing roles, understanding how enjoyment influences satisfaction is vital for successful implementation and adoption. Several studies found a positive link between enjoyment and satisfaction in the context of AI and technology-mediated services. For instance, [Chiang et al. \(2021\)](#) examined social shopping behaviors and found that enjoyment is vital in influencing customer satisfaction and loyalty in online environments. Similarly, [Rajaobelina et al. \(2021\)](#) investigated customer interactions with AI chatbots, revealing that enjoyment is positively associated with satisfaction and loyalty. Their finding highlights the necessity of designing chatbot interactions that are functional and enjoyable for users.

[Belanche et al. \(2021\)](#) introduced the HVL model to evaluate customer reactions to frontline service robots in the hospitality sector. Their findings indicated that perceived warmth—related to enjoyment—positively influenced emotional value expectations and subsequent satisfaction, which suggests that creating enjoyable interactions with AI-based services can significantly enhance the overall customer experience in hospitality settings. Supporting this notion, [Lee and Peng \(2021\)](#) studied environmental leisure farms and found that experiential value, including enjoyment, positively affected satisfaction. Further research emphasized the importance of enjoyment in driving satisfaction with AI services, aligning with broader technology adoption theories. For instance, [Hsu and Lin \(2023\)](#) examined user satisfaction with customer service chatbots and found that conversational quality, which

includes enjoyment, significantly influenced satisfaction. Similarly, [Hernandez-Ortega and Ferreira \(2021\)](#) explored intelligent voice assistants and concluded that positive experiences, including enjoyment, increase satisfaction and loyalty. They stressed the need to design AI interactions that evoke positive emotions and enjoyment.

Flow experience, closely related to enjoyment, is also connected to satisfaction with AI-based services. [Wei et al. \(2021\)](#) found that flow experiences positively influenced satisfaction and loyalty in AI-powered recommendation systems. [Meeprom and Suttikun \(2024\)](#) supported this idea, discovering that customer engagement with AI-based services—particularly enjoyment elements—positively impacted satisfaction in the hospitality industry. Overall, the positive influence of enjoyment on satisfaction with AI-based services aligns with the broader theories of technology adoption and user experience. TAM posits that perceived enjoyment is a crucial factor affecting user attitudes and behavioral intentions ([Davis et al., 1992](#)). Additionally, the S-O-R framework suggests that enjoyable stimuli (AI interactions) generate positive emotional responses (satisfaction) and behavioral outcomes (loyalty). While [Pizzi et al. \(2023\)](#) noted that privacy concerns could moderate the relationship between enjoyment and satisfaction, the prevailing evidence suggests a positive correlation between enjoyment and satisfaction with AI-based customer services in the hospitality industry. Given these findings, we present the following hypothesis.

H5. Enjoyment positively influences satisfaction with AI-based customer services.

The relationship between customer satisfaction and loyalty is well-established in the marketing literature, particularly concerning AI-based services and the hospitality industry. As companies increasingly adopt AI technologies (such as chatbots and virtual assistants) to enhance customer service, understanding how satisfaction with such services affects customer loyalty becomes crucial. Several studies have shown that satisfaction with AI-based customer services can positively influence customer loyalty across various sectors, including hospitality. For instance, [Meeprom et al. \(2024\)](#) examined how AI and employee-based customer services affected customer engagement and loyalty during the COVID-19 pandemic. They revealed that customer engagement with AI-based services significantly boosted loyalty. Similarly, [Chen et al. \(2023\)](#) investigated the influence of AI chatbots on customer loyalty in the home-sharing industry, concluding that customer trust and engagement with AI chatbots increased loyalty intentions. Additional research has demonstrated positive links between satisfaction with AI services and customer loyalty across industries. For example, [Hsu and Lin \(2023\)](#) found that satisfaction with AI-chatbot services significantly influences customer loyalty, emphasizing the importance of core service quality in driving loyalty toward AI-powered chatbots. Niu and Hayat (2024) also examined loyalty toward AI chatbots and identified satisfaction as a critical driver of loyalty and repurchase intentions.

The extant literature has also explored the mechanisms through which AI service satisfaction impacts loyalty. For example, [Wu et al. \(2021\)](#) proposed that emotional factors play a crucial role, showing that positive emotions mediate the relationship between AI service quality and customer loyalty. Additionally, [Hernandez-Ortega and Ferreira \(2021\)](#) highlighted the importance of emotional connections, demonstrating that consumers could develop feelings for smart voice assistants, increasing service loyalty. Furthermore, several studies have identified specific aspects of AI-enabled services contributing to satisfaction and loyalty. In the hospitality industry, [Holendova et al. \(2024\)](#) found that the personalization capabilities of AI chatbots significantly affected customer satisfaction and loyalty intentions. Additionally, [Chiang et al. \(2021\)](#) emphasized personalization as a driver of loyalty in AI-powered hotel services. Similarly, [Hasan et al. \(2021\)](#) demonstrated that consumer trust is another essential mediator between AI service satisfaction and loyalty, finding that trust in voice-controlled AI positively influences brand loyalty.

Furthermore, several studies in the hospitality sector have reinforced

the connection between AI service satisfaction and loyalty. Liu et al. (2024) found that AI-enabled personalization and hotel service quality positively affect customer satisfaction and loyalty intentions. Al-Hyari et al. (2023) demonstrated that customer satisfaction with AI-powered hotel services increased loyalty and positive word-of-mouth behaviors. These findings suggest hospitality firms can leverage AI technologies to create satisfying service experiences that translate into stronger customer loyalty. While challenges exist in implementing these technologies, the overall evidence indicates that AI-enabled services can enhance customer satisfaction and drive loyalty among hospitality customers when executed effectively. In light of these findings, we present the following hypothesis.

H6. Satisfaction with AI-based customer services positively influences customer loyalty.

Fig. 1 presents the research framework.

3. Research methodology

This research used quantitative and inductive reasoning to examine how AI-enabled service attributes—reliability (REL), perceived intelligence (PIG), service efficiency (SER), aesthetics (AES), and enjoyment (ENJ)—influence satisfaction with AI-based customer services (SAT), which in turn affects customer loyalty (LOY) in the context of chain hotels. REL and SER represent the utilitarian aspects of AI services, while PIG highlights the unique characteristics that differentiate AI services

from traditional ones. AES and ENJ capture the hedonic aspects of these services. SAT mediates the relationship between AI service attributes and LOY in this framework. This study used a cross-sectional research design that integrates elements from the TAM, SERVQUAL, S-O-R, and HVL models to comprehensively understand how AI-enabled services impact customer loyalty in chain hotels. We define the sample, create the research instrument, and develop a data collection and analysis plan with the details outlined below.

3.1. Population and sample

This study employed a rigorous methodological purposive sampling approach with a strategic focus on geography to investigate AI-enabled services in the hospitality sector. The selection process demanded specific inclusion criteria: Participants must have directly engaged with AI chatbot services in chain hotels within the last six months and stayed at least one night. These criteria ensured that the study participants had recent and substantial experience, allowing them to provide informed assessments of their interactions with AI services (Chen et al., 2023; Hsu & Lin, 2023).

The selection of coastal tourism destinations in Western Thailand serves as a strategic factor that justified the study area. This region boasts a high concentration of international chain hotels, with 47 properties implementing AI chatbot services, creating an ideal environment for studying AI-enabled customer services (Hansasooksin et al., 2024). This area also attracts diverse guest demographics including both

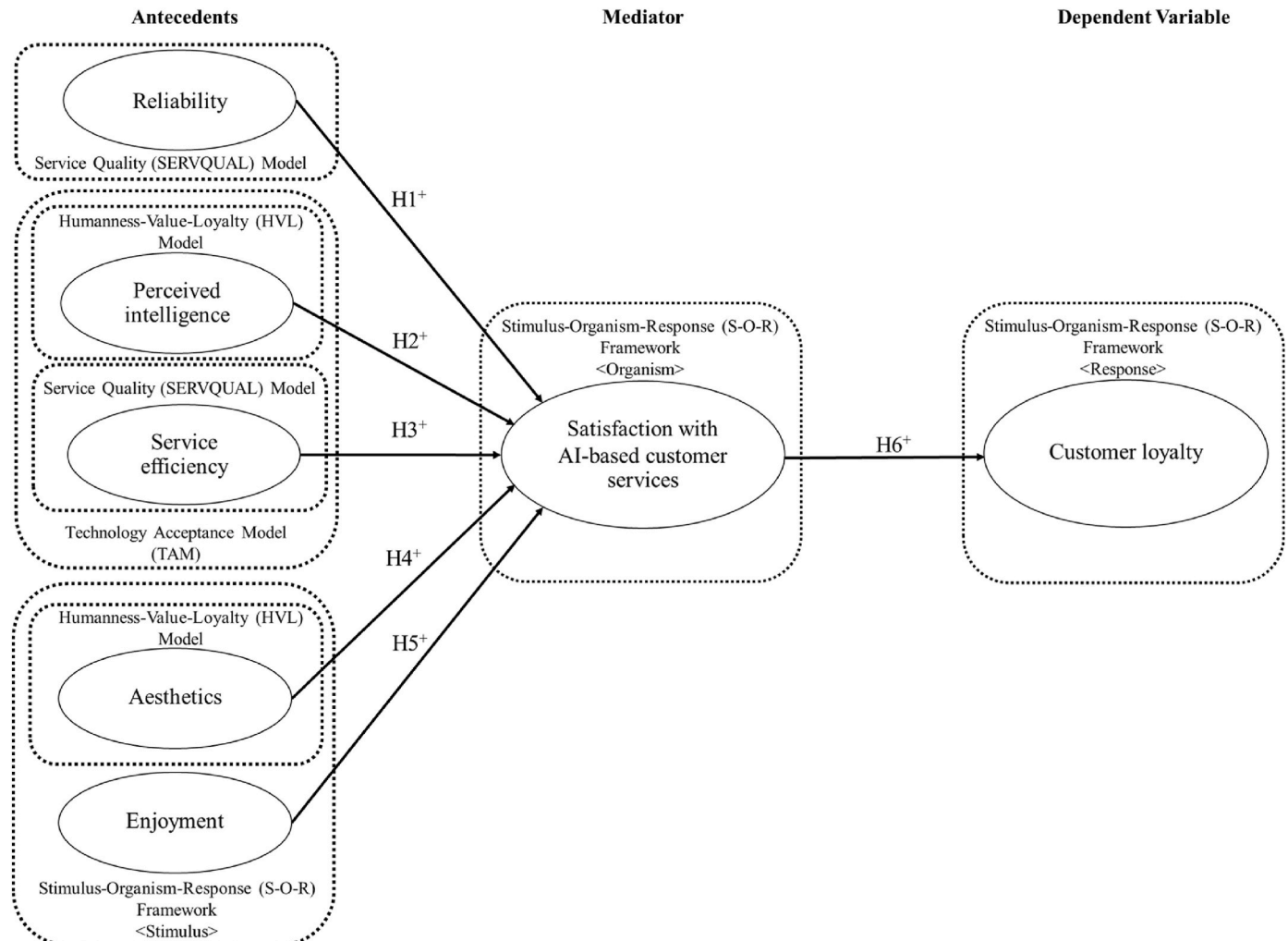


Fig. 1. Research framework.

domestic and international travelers who offer varied perspectives on AI service interactions (Goffi et al., 2021). The distinct peak and off-peak seasonal patterns of this region also provide an opportunity to examine AI service performance under different operational conditions (Liu et al., 2024).

The sample size was established following statistical guidelines and theoretical recommendations. Nunnally's (1967) criterion of a 10:1 respondent-to-variable ratio suggested a minimum sample size of 220 respondents based on the 22 observed variables, while Kline (1998) recommended that a sample size between 100 and 200 respondents was adequate for structural equation modeling studies. To further validate these recommendations, we checked the appropriate sample size using Free Statistics Calculators Version 4.0. This program included an anticipated effect size of 0.3 (medium level) which was deemed suitable for social science research examining consumer behavior because relationships that are too high or too low may not accurately reflect reality. We set the desired statistical power level at 0.8 (80%) for the standard research threshold, indicating a 20% chance of a Type II error – failing to reject a false null hypothesis (danielsoper.com, 2024). Our results indicated that a minimum sample size of 180 respondents was suitable.

After data collection, 201 questionnaires were returned. The study team assessed the completed questionnaires and found that 186 were legitimate for data analysis. The final sample size of 186 participants demonstrated statistical and theoretical robustness through multiple validation approaches. Nunnally's (1967) 10:1 criterion suggested 220 respondents but our sample size aligned with Kline's (1998) parameter of 100–200 respondents for structural equation modeling research. Statistical power analysis confirmed the sample's adequacy, indicating robust power (0.95) for detecting medium effect sizes ($f^2 = 0.15$) at $\alpha = 0.05$.

Comparative analyses by recent studies in hospitality AI research further supported the adequacy of the sample size. Notable studies included Thakur et al. (2025) with 102 participants, Yin et al. (2024) with 103 participants, and Zhang and Jin (2023) with 100 participants—all generating significant findings with comparable sample sizes. The calculations for our study incorporated an anticipated medium effect size (0.3) and statistical power (0.8) using Free Statistics Calculators Version 4.0, adhering to established parameters in social science research.

The demographic composition of the sample, particularly the predominance of participants aged 20–30 (76.3%), enhances the relevance of the findings to better understand patterns in AI technology adoption. This age group represents early adopters, whose usage patterns often indicate emerging market trends. Our findings primarily apply to similar tourism contexts in emerging markets, and this focused approach provides precise, contextually grounded insights that may not arise from geographically dispersed sampling.

Overall, this comprehensive sampling methodology strengthens internal validity by systematic participant screening and external validity through alignment with existing research. Future studies can expand on our findings by replicating the research in diverse geographic and demographic contexts, thereby building on the methodological foundation established in this study.

3.2. Research instrument

This study's research instrument was a carefully designed questionnaire developed through a thorough adaptation and validation process. The questionnaire comprised two main sections: a demographic profile with 7 items and a 22-item scale assessing how AI-enabled service attributes influence satisfaction with AI-based customer services, which impacts customer loyalty in the chain hotel context. REL was measured using a scale adapted from the extant literature (Belanche et al., 2021; Chen et al., 2023; Hasan et al., 2021; Hsu & Lin, 2023; Noor et al., 2024). PIQ was measured using a scale adapted from several studies (Belanche

et al., 2021; Chen et al., 2023; Hsu & Lin, 2023; Mustak et al., 2021; Noor et al., 2024). SER was assessed using a scale adapted from the extant research (Belanche et al., 2021; Chen et al., 2023; Hsu & Lin, 2023; Mustak et al., 2021; Plattfaut & Koch, 2021). AES was measured using a scale adapted from the relevant research (Belanche et al., 2021; Chen et al., 2021; Reig et al., 2021). ENJ was assessed using a scale adapted from the literature (Belanche et al., 2021; Chiang et al., 2021; Hernandez-Ortega & Ferreira, 2021; Huh et al., 2024; Jeon, 2024). SAT was measured using a scale adapted from existing studies (Belanche et al., 2021; Chen et al., 2023; Chiang et al., 2021; Hernandez-Ortega & Ferreira, 2021; Hsu & Lin, 2023; Noor et al., 2024; Prentice et al., 2022; Rajaobelina et al., 2021). LOY was assessed with a scale adapted from the existing research (Belanche et al., 2021; Chen et al., 2023; Chiang et al., 2021; Hasan et al., 2021; Hernandez-Ortega & Ferreira, 2021; Hsu & Lin, 2023; Jeon, 2024; Noor et al., 2024; Prentice et al., 2022; Rajaobelina et al., 2021).

We used a 7-point Likert scale to capture detailed responses, allowing for a more nuanced analysis of participant attitudes and perceptions; the rating scale ranged from 1 (strongly disagree) to 7 (strongly agree). The instrument underwent a thorough validation process, starting with a content validity assessment conducted by three education experts using the index of item-objective congruence (IOC) method (Rovinelli & Hambleton, 1977). All items showed satisfactory concordance, with IOC values ranging from 0.67 to 1.00, surpassing the threshold of 0.50. A pilot test was then administered with 30 participants to evaluate the instrument's reliability. Cronbach's alpha coefficients were calculated for each construct, resulting in values between 0.800 and 0.893, well above the recommended threshold of 0.7 (Hair et al., 2010). This rigorous validation process, including content validity and internal consistency reliability, enhances the credibility and trustworthiness of the collected data. The final refined questionnaire was distributed to a sample of 180 participants, selected through purposive sampling through face-to-face interactions to ensure representativeness. This methodological approach adheres to the best practices in survey research and provides a solid foundation for subsequent statistical analyses and interpretations.

3.3. Data collection

This research collected data from a sample group that had experience using AI-chatbot customer services in chain hotel businesses. A face-to-face questionnaire (the research instrument) was administered in Thailand's western coastal tourism region, and purposive sampling was utilized from August to September 2024. The questionnaire's first section included two screening questions. The first question asked whether the respondent agreed to provide information by completing the questionnaire. The second question determined whether the respondents had experience using AI-chatbot customer services in chain hotels, which ensured that the respondents had genuine experience with the technology. At the end of the data collection period, 201 questionnaires were returned. The research team examined the completeness of the questionnaires and determined that 186 were valid for data analysis. The decision to use partial least squares structural equation modeling (PLS-SEM) was based on the sample size, which exceeded the threshold of 100 participants recommended by Reinartz et al. (2009).

3.4. Data analysis

The research team verified the questionnaires' completeness before coding and processing the data using computer software (SPSS) for statistical analysis. Descriptive statistics, including the frequency distribution and percentage calculations, were utilized to present the sample's demographic characteristics. We employed the PLS-SEM methodology to conduct hypothesis testing, which is effective for analyzing the collected empirical data. Hair et al. (2011) indicated that the PLS-SEM methodology is appropriate for research to assess the

predictive capacity of complex models involving multiple dependent and independent variables and moderator and mediator variables—especially in contexts with small sample sizes. Given this study's conditions, the use of PLS-SEM is justified. The data analysis process followed a two-phase approach—the first step involved evaluating the validity and reliability of the measurement model. In the second phase, the structural model was assessed to determine the statistical significance of the proposed hypotheses. The validity and reliability of the measurement model were evaluated through several tests, including convergent validity (outer loadings >0.70 and average variance extracted >0.50), internal consistency (Cronbach's alpha >0.70 and composite reliability >0.70), and discriminant validity (using the Fornell–Larcker criterion and the heterotrait-monotrait ratio of correlations). Once the validity and reliability of the measurement model were established, the structural model was evaluated based on the magnitude and significance of the path coefficients and the coefficient of determination (R^2).

4. Results

The researchers used IBM SPSS 27.0 to assess the common method bias (CMB) and demographics; Smart PLS 4 software was used to evaluate the measurement and structural model techniques. Data were analyzed in two stages: (1) model fit, reliability, and validity, and (2) hypothesis testing.

4.1. Common method bias (CMB)

CMB arises from instrument rather than respondent biases. CMB must be assessed when data are collected using a single instrument and self-reported surveys (Podsakoff et al., 2012; Williams & McGonagle, 2016). It is important to ensure that the questions for the predictor and criterion variables are identical for each participant. Harman's single-factor methodology can be used to evaluate the CMB. This approach examines the distribution of variance percentages and involves an exploratory factor analysis of all items. Ideally, the first unrotated factor should account for less than 50% of the total variance (Podsakoff et al., 2012). The CMB is assessed through Harman's single-factor score, aggregating all latent variables into one component. CMB does not significantly affect the data and outcomes if this first factor accounts for less than 50% of the variance. In this study, the total variance extracted from the initial Eigenvalues revealed a first-factor loading of 49.27%, which is below 50%; thus, we conclude that this dataset does not exhibit CBM (Podsakoff et al., 2012).

4.2. Respondents' profiles

The study sample comprised 186 participants, predominantly female (56.5%), with males representing 24.7% and LGBTQIA + individuals accounting for 16.1%. Most respondents were young single (82.8%) adults aged 20 to 30 (76.3%). Educational achievement was notably high, with 74.2% holding a bachelor's degree and 8.1% completing postgraduate studies. Regarding employment, nearly half of the participants (49.5%) worked in private companies, while 26.9% were students. Income levels varied; 28.5% earned less than 10,000 Thai baht (THB) monthly, and 27.4% earned between 10,000 and 20,000 THB. Notably, 21% reported monthly incomes between 20,001 and 30,000 THB. The primary reasons respondents visited chain hotels were relaxation and leisure (83.3%), followed by dining (5.4%) and attending events (4.3%). Table 1 provides further details about the respondents' profiles.

4.3. Measurement model evaluation

This study's measurement model evaluation demonstrates robust validity and reliability across all constructs, adhering to the criteria

Table 1

Respondent's profile (N = 186).

Demographics	Distribution	Frequency	Percent
Gender	Male	46	24.7
	Female	105	56.5
	LGBTQIA+	30	16.1
	Prefer not to say	5	2.7
Age	20–30 years	142	76.3
	31–40 years	21	11.3
	41–50 years	12	6.5
	51 years and above	11	5.9
Marital status	Single	154	82.8
	Married without children	9	4.8
	Married with children	19	10.2
	Divorced	2	1.1
	Separated due to work requirements	2	1.1
Education level	Below Bachelor's degree	33	17.7
	Bachelor's degree	138	74.2
	Higher than Bachelor's degree	15	8.1
Occupation	Student	50	26.9
	Government officer	7	3.8
	Company employee	92	49.5
	Business owner or Self-employed	21	11.3
	Freelance	7	3.8
	Agriculturist	3	1.6
	Retired	6	3.2
Income (Baht/Month)	Less than 10,000	53	28.5
	10,000–20,000	51	27.4
	20,001–30,000	39	21.0
	30,001–40,000	17	9.1
	40,001–50,000	9	4.8
	50,000 and above	17	9.1
Purpose of chain hotel visit or service use	For relaxation or leisure	155	83.3
	For meetings or seminars	7	3.8
	To use restaurant services	10	5.4
	To use spa services	6	3.2
	To attend events	8	4.3

established by Sarstedt et al. (2014). Convergent validity is strongly supported by the outer loadings of all items, which range from 0.781 to 0.926, comfortably exceeding the recommended threshold of 0.70 (Sarstedt et al., 2014). All constructs' average variance extracted (AVE) values surpass the 0.50 benchmark from 0.690 to 0.824, further reinforcing convergent validity (Sarstedt et al., 2014). The model also exhibits excellent internal consistency, as evidenced by Cronbach's alpha values ranging from 0.801 to 0.893 and composite reliability (CR) scores between 0.804 and 0.893, well above the 0.70 criterion (Sarstedt et al., 2014). These comprehensive results indicate the measurement model's high validity and reliability, suggesting that the constructs are well-defined and consistently measured across all items. The strong performance across these multiple indicators provides a solid foundation for the research model, reinforcing the measurements' accuracy and consistency (Sarstedt et al., 2014). Table 2 shows that all constructs met the criterion for good reliability and validity.

4.4. Discriminant validity

The discriminant validity assessment is crucial in evaluating the measurement model's quality in PLS-SEM. The Fornell–Larcker criterion proposed by Bagozzi and Yi (1989) offers a robust method for this assessment. This criterion establishes discriminant validity when the AVE square root for each construct exceeds its correlations with other constructs in the model. In Table 3, the diagonal elements (in bold) represent the square roots of the AVE for each construct, while the off-diagonal elements show the correlations between constructs. The diagonal value (square root of AVE) is more significant for each construct than its corresponding row and column values (correlations

Table 2
Construct reliability and validity.

Construct	Codes	Items	VIF	FL	α	CR	AVE	Source
Reliability (REL)	REL1	The AI-Chatbot customer services of the chain hotel you have used functions correctly on the first attempt.	2.276	0.860	0.849	0.853	0.690	Belanche et al. (2021), Chen et al. (2023), Noor et al. (2024), Hasan et al. (2021), and Hsu and Lin (2023)
	REL2	The AI-Chatbot customer service of the chain hotel you have used operates continuously.	2.466	0.880				
	REL3	The AI-Chatbot customer services of the chain hotel you have used have no data leakage.	1.657	0.781				
	REL4	The AI-Chatbot customer service of the chain hotel you used is modern.	1.784	0.797				
Perceived intelligence (PIG)	PIG1	The AI-Chatbot customer services of the chain hotel you have used displays results quickly.	1.624	0.845	0.801	0.804	0.715	Belanche et al. (2021), Chen et al. (2023), Mustak et al. (2021), Noor et al. (2024), and Hsu and Lin (2023)
	PIG2	The AI-Chatbot customer services of the chain hotel you have used can verify various privileges.	1.801	0.855				
	PIG3	The AI-Chatbot customer services of the chain hotel you have used can be customized according to user needs.	1.754	0.835				
Service efficiency (SER)	SER1	The AI-Chatbot customer services of the chain hotel you have used are not complex.	2.195	0.887	0.846	0.849	0.764	Belanche et al. (2021), Chen et al. (2023), Mustak et al. (2021), Hsu and Lin (2023), and Plattfauf and Koch (2021)
	SER2	The AI-Chatbot customer service of the chain hotel you used can provide accurate information.	2.185	0.891				
	SER3	The AI-Chatbot customer services of the chain hotel you have used do not require much effort for first-time use.	1.831	0.844				
Aesthetics (AES)	AES1	The AI-Chatbot customer service of the chain hotel you used has a modern screen design.	1.945	0.857	0.853	0.858	0.773	Belanche et al. (2021), Chen et al. (2021), and Reig et al. (2021)
	AES2	The AI-Chatbot customer services of the chain hotel you have used has a screen design that presents clear and complete content.	2.285	0.888				
	AES3	The AI-Chatbot customer services of the chain hotel you have used has an attractive and user-friendly screen design.	2.168	0.892				
Enjoyment (ENJ)	ENJ1	You find entertainment from using the AI-Chatbot customer services of the chain hotel you have used.	2.913	0.911	0.893	0.893	0.824	Belanche et al. (2021), Chiang et al. (2021), Hernandez-Ortega and Ferreira (2021), Huh et al. (2024), and Jeon (2024)
	ENJ2	You enjoy using the AI-Chatbot customer services of the chain hotel you have used.	2.271	0.885				
	ENJ3	You will have fun using the AI-Chatbot customer service of the chain hotel you used.	3.304	0.926				
Satisfaction with AI-based customer services (SAT)	SAT1	You are satisfied with all aspects of using the AI-Chatbot customer services of the chain hotel you have used.	2.111	0.875	0.868	0.869	0.791	Belanche et al. (2021), Chen et al. (2023), Chiang et al. (2021), Hernandez-Ortega and Ferreira (2021), Hsu and Lin (2023), Noor et al. (2024), Prentice et al. (2022), and Rajaobelina et al. (2021)
	SAT2	You have a positive attitude toward using the AI-Chatbot customer services of the chain hotel you have used.	2.288	0.893				
	SAT3	Overall, you are satisfied with using the AI-Chatbot customer services of the chain hotel you have used.	2.500	0.900				
Customer loyalty (LOY)	LOY1	You will share positive experiences or impressive stories from using the AI-Chatbot customer services of the chain hotel you have used with friends or acquaintances.	2.572	0.907	0.858	0.859	0.779	Belanche et al. (2021), Chen et al. (2023), Chiang et al. (2021), Hasan et al. (2021), Hernandez-Ortega and Ferreira (2021), Hsu and Lin (2023), Jeon (2024), Noor et al. (2024), Prentice et al. (2022), and Rajaobelina et al. (2021)
	LOY2	You will use the AI-Chatbot customer services of the chain hotel you have used again if you can stay or use services at a chain hotel.	2.212	0.883				
	LOY3	You will use the AI-Chatbot customer services of the chain hotel you have used as your first choice if you can stay or use services at a chain hotel.	1.949	0.857				

Note: VIF = Variance Inflation factor; factor Loadings; α = Cronbach's alpha; CR = Composite reliability; AVE = Average variance extracted.

Table 3
Discriminant validity for the Fornell–Larcker criterion.

Construct	PIG	ENJ	LOY	SER	REL	SAT	AES
Perceived intelligence (PIG)	0.845						
Enjoyment (ENJ)	0.668	0.908					
Customer loyalty (LOY)	0.750	0.810	0.883				
Service efficiency (SER)	0.781	0.666	0.707	0.874			
Reliability (REL)	0.757	0.695	0.748	0.713	0.830		
Satisfaction with AI-based customer services (SAT)	0.723	0.825	0.833	0.750	0.678	0.889	
Aesthetics (AES)	0.728	0.697	0.746	0.720	0.715	0.719	0.879

Note: The diagonal values in bold are the square root of the AVE values.

with other constructs). For instance, the perceived intelligence (PIG) construct has a square root of AVE of 0.845, higher than its correlations with other constructs (ranging from 0.668 to 0.781). This pattern holds for all constructs in the model. This consistent observation across all constructs indicates that the Fornell–Larcker criterion is satisfied, confirming the discriminant validity (Bagozzi & Yi, 1989). This result suggests that each construct in the model is genuinely distinct from other constructs, capturing phenomena not represented by other constructs in the model. This differentiation is crucial for ensuring that the constructs measure what they aim to measure, thus strengthening the overall validity of the research model and its findings.

4.5. Structural model evaluation and hypotheses testing

Assessing the collinearity in the structural model is a crucial step in ensuring the validity of the results. This study used the variance Inflation factor (VIF) to examine potential multicollinearity issues among the latent variables of the predictor constructs. Hair et al. (2019, 2020) suggested that VIF values greater than 3 indicate potential collinearity

problems, with values between 3 and 5 signaling potential collinearity issues. The VIF values in Table 1 show that the VIF values range from 1.624 (PIG1) to 3.304 (ENJ3), with most falling comfortably below 3. Only one indicator, ENJ3, slightly exceeds the lower bound of the cautionary range, with a VIF of 3.304. While worth noting, this isolated case does not suggest widespread collinearity issues within the model. The predominantly low VIF values indicate that multicollinearity is not a significant concern in this structural model, suggesting that the predictor variables are sufficiently distinct, allowing for a reliable estimation of their effects on the dependent variables. The absence of substantial collinearity enhances the stability and interpretability of the model’s results, providing a solid foundation for further analysis and interpretation of the structural relationships (Hair et al., 2019, 2020).

After validating the measurement model, we conducted a structural model analysis utilizing a bootstrapping resampling approach with 5000 iterations; this approach allowed us to test the hypotheses and determine the statistical significance of the model coefficients (Streukens & Leroi-Werelds, 2016). Table 3 and Fig. 2 present the empirical results, including the path coefficients (β), t-statistics, and p-values for each

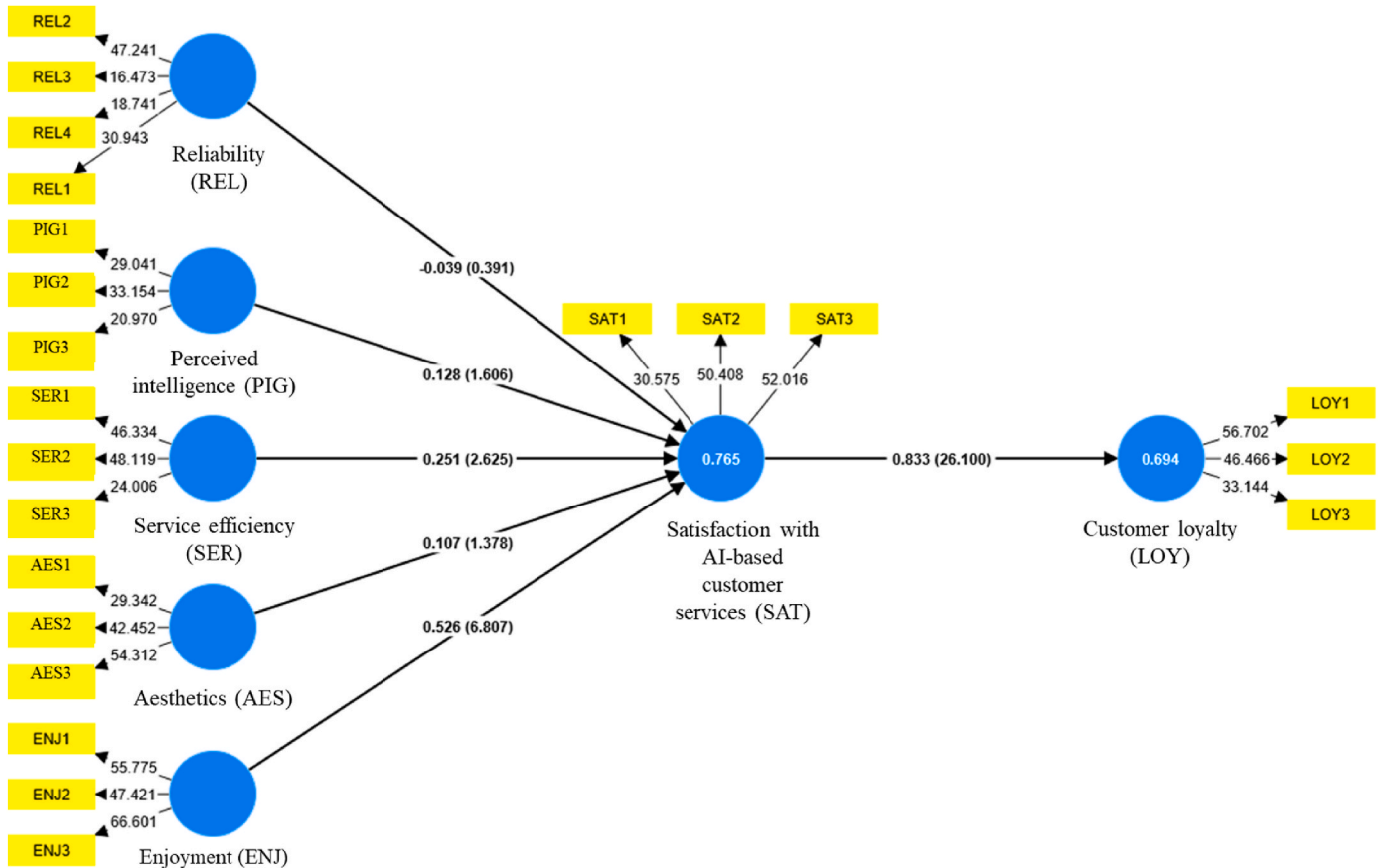


Fig. 2. The statistical results of our study model.

hypothesized relationship. Following Hair et al. (2014), where t-values of 1.96 or higher indicate statistically significant relationships between constructs, the results show that three of the six hypothesized relationships are statistically significant.

1. Service efficiency (SER) → Satisfaction with AI-based customer services (SAT) ($\beta = 0.251$, $t = 2.625$, $p = 0.009$)
2. Enjoyment (ENJ) → Satisfaction with AI-based customer services (SAT) ($\beta = 0.526$, $t = 6.807$, $p = 0.000$)
3. Satisfaction with AI-based customer services (SAT) → Customer loyalty (LOY) ($\beta = 0.833$, $t = 26.100$, $p = 0.000$)

These significant relationships indicate that SER and ENJ positively influence SAT, while SAT strongly impacts LOY. The relationship between SAT and LOY is particularly robust, as demonstrated by the high path coefficient and t-value.

The relationships between REL and SAT ($\beta = -0.039$, $t = 0.391$), PIG and SAT ($\beta = 0.128$, $t = 1.606$), and AES and SAT ($\beta = 0.107$, $t = 1.378$) did not reach the significance threshold. This result indicates that the empirical data do not support these proposed relationships.

Hair et al. (2019) proposed that the coefficient of determination (R^2) for the endogenous variables offers insight into the model's predictive power. The R^2 value for SAT is 0.765, indicating that the exogenous variables explain 76.5% of the variance in satisfaction. Meanwhile, the R^2 for LOY is 0.694, suggesting that satisfaction accounts for 69.4% of the variance in LOY.

The results from the PLS-SEM analysis provide valuable insights into the relationships between the constructs and the overall predictive power of the model. These findings highlight the significance of service efficiency and enjoyment in driving customer satisfaction, which strongly influences customer loyalty; however, the model shows that reliability, perceived intelligence, and aesthetic design do not significantly affect satisfaction.

Based on the structural model evaluation and the testing of the hypotheses, Table 4 and Fig. 2 summarize the results of the research hypothesis testing.

4.6. Mediation analysis

Examining the mediating effects across AI service attributes revealed varying mechanisms that influenced customer loyalty. Using bootstrapping procedures with 5000 iterations in PLS-SEM allowed a comprehensive analysis of direct and indirect effects between service attributes and loyalty outcomes, highlighting distinct mediation patterns that warranted careful consideration.

Statistical analysis identified two significant mediation pathways of theoretical and practical importance. First, satisfaction was a potent mediator between enjoyment and customer loyalty ($\beta = 0.438$, $t = 6.623$, $p = 0.000$). This finding provided empirical evidence that positive emotional experiences with AI services foster enduring loyalty by enhancing satisfaction. Second, satisfaction also significantly mediated the relationship between service efficiency and loyalty ($\beta = 0.209$, $t =$

2.680, $p = 0.007$), indicating that operational excellence in AI service delivery increases loyalty through improved customer satisfaction.

The comparative magnitudes of these mediating effects deserve particular attention. The more substantial indirect effect associated with enjoyment ($\beta = 0.438$) compared to service efficiency ($\beta = 0.209$) underscores the importance of emotional engagement over functional performance in fostering loyalty through satisfaction. This finding aligns with literature in the hospitality sector, emphasizing the role of experiential factors in guest satisfaction and loyalty formation (Hernandez-Ortega & Ferreira, 2021; Meeprom & Suttikun, 2024).

Further analyses showed that satisfaction did not significantly mediate the relationships between reliability ($\beta = -0.033$, $t = 0.391$, $p = 0.696$), aesthetics ($\beta = 0.089$, $t = 1.360$, $p = 0.174$), or perceived intelligence ($\beta = 0.106$, $t = 1.590$, $p = 0.112$) and loyalty. Rather than indicating methodological limitations, these findings highlighted the complex nature of loyalty formation in AI-enabled services, suggesting that while reliability, aesthetics, and perceived intelligence are theoretically important constructs, their influence on loyalty may occur through mechanisms other than satisfaction.

This selective mediation pattern enhances our theoretical understanding by demonstrating the non-uniformity of the mediating role of satisfaction across different AI service attributes. Our findings established satisfaction as a critical mediating mechanism for service aspects related to efficiency and enjoyment, while suggesting that other theoretical pathways may better explain how reliability, aesthetics, and perceived intelligence impact loyalty outcomes. This nuanced understanding offers empirical guidance for strategic resource allocation in designing and implementing AI services.

The differential mediation effects observed in this analysis carry significant implications for theoretical development and managerial practice. Theoretically, these findings suggest a need for more complex models, incorporating multiple mediating mechanisms, to fully elucidate the relationships between AI service attributes and customer loyalty. For practitioners, the results indicate that initiatives aimed at fostering loyalty through satisfaction should prioritize the development of enjoyable and efficient AI service interactions while exploring alternative pathways to leverage the benefits of reliability, aesthetics, and perceived intelligence. Table 5 presents a detailed analysis of the effects of mediation.

4.7. Model explanatory power

The model's explanatory power was evaluated using the coefficient of determination (R^2) recommended by Bagozzi and Yi (1989). The results show that all R^2 scores surpassed the recommended threshold of 0.10, indicating the model's predictive accuracy. Specifically, the R^2 values for LOY and SAT were 0.694 and 0.765, respectively. According to Henseler et al. (2009), these values can be interpreted as moderate to substantial, with R^2 values of 0.75 considered substantial, 0.50 moderate, and 0.25 weak. This outcome implies that the model accounts for a significant portion of the variance in LOY and SAT, reflecting strong predictive power. Table 6 presents the quality criteria.

The goodness of fit for the structural model was assessed using the standardized root mean square residual (SRMR) criterion. The estimated model achieved an SRMR value of 0.068, below the recommended threshold of 0.080 (Putnick & Bornstein, 2016). This result indicates

Table 4
Results of the hypothesis testing.

Hypothesis	Relationships	Original sample (β)	T statistics	P values	Decision
1	REL → SAT	−0.039	0.391	0.696	Not support
2	PIG → SAT	0.128	1.606	0.108	Not support
3	SER → SAT	0.251	2.625	0.009	Support
4	AES → SAT	0.107	1.378	0.168	Not support
5	ENJ → SAT	0.526	6.807	0.000	Support
6	SAT → LOY	0.833	26.100	0.000	Support

Table 5
Mediation effects.

Relationships	Original sample (β)	T statistics	P values	Decision
REL → SAT → LOY	−0.033	0.391	0.696	No
UID → SAT → LOY	0.089	1.360	0.174	No
COM → SAT → LOY	0.106	1.590	0.112	No
ENJ → SAT → LOY	0.438	6.623	0.000	Yes
PER → SAT → LOY	0.209	2.680	0.007	Yes

Table 6
Quality criteria.

Construct	R-square	R-square adjusted
LOY	0.694	0.692
SAT	0.765	0.758

that the model fits well and adequately meets the study’s objectives. [Table 7](#) presents the model fit value.

We examined f^2 values to assess the impact of the independent constructs on the dependent constructs. [Hair et al. \(2020\)](#) indicated that f^2 values of 0.02, 0.15, and 0.35 correspond to small, medium, and large effects, respectively. The results reveal varying effect sizes.

1. Satisfaction with AI-based customer services (SAT) → Customer loyalty (LOY) ($f^2 = 2.268$): This indicates a substantial effect of SAT on LOY.
2. Enjoyment (ENJ) → Satisfaction with AI-based customer services (SAT) ($f^2 = 0.486$): ENJ has a large effect on SAT.
3. Service efficiency (SER) → Satisfaction with AI-based customer services (SAT) ($f^2 = 0.086$): The perceived ease of use has a small to medium effect on SAT.
4. Perceived intelligence (PIG) → Satisfaction with AI-based customer services (SAT) ($f^2 = 0.020$), aesthetics (AES) → Satisfaction with AI-based customer services (SAT) ($f^2 = 0.017$), and Reliability (REL) → Satisfaction with AI-based customer services (SAT) ($f^2 = 0.002$): These constructs show small or negligible effects on SAT.

These findings offer valuable insights into the relative importance of different predictor variables in the structural model. The results underscore the crucial role of SAT in predicting LOY, along with ENJ’s significant influence on SAT. SER also has a notable, albeit smaller, effect on SAT; however, the effects of PIG, AES, and REL on SAT are minimal in this context. [Table 8](#) presents the effect sizes of the independent constructs on the dependent constructs.

4.8. Model’s predictive power

The predictive value of the model was assessed using cross-validated redundancy analysis (Q^2), employing the blindfolding strategy as recommended by [Hair et al. \(2014\)](#) and [Ittefaq et al. \(2024\)](#). According to [Shmueli et al. \(2019\)](#), a model is considered predictive when the Q^2 value for the dependent variable exceeds zero. In this study, the Q^2 values for LOY and SAT are notably above zero, with values of 0.706 and 0.738, respectively ([Table 9](#)). These results indicate that the external constructs effectively predict the endogenous constructs being analyzed. [Geisser \(1974\)](#), [Chin \(2010\)](#), and [Hair et al. \(2017\)](#) suggest that Q^2 values of 0.02, 0.15, and 0.35 represent small, medium, and large predictive relevance, respectively. In this context, the Q^2 values for both LOY and SAT far exceed 0.35, indicating the considerable predictive importance of the external constructs for these endogenous variables. When considered alongside the previously reported R^2 values (0.694 for LOY and 0.765 for SAT), the magnitudes of these Q^2 values provide a comprehensive assessment of the model’s prediction accuracy, as [Geisser \(1974\)](#) suggested. The high Q^2 values complement the strong R^2 values, reinforcing the model’s robust predictive capability. Additionally, the root mean square error (RMSE) and mean absolute error (MAE) values reported in [Table 9](#) provide further insight into the model’s predictive accuracy. The relatively low RMSE (0.549 for LOY and 0.520 for SAT) and MAE (0.409 for LOY and 0.360 for SAT) suggest good

Table 7
Model fit.

Criteria	Saturated model	Estimated model
SRMR	0.049	0.068

Table 8
f-square.

Construct	PIG	ENJ	LOY	SER	REL	SAT	AES
Perceived intelligence (PIG)						0.020	
Enjoyment (ENJ)						0.486	
Customer loyalty (LOY)							
Service efficiency (SER)						0.086	
Reliability (REL)						0.002	
Satisfaction with AI-based customer services (SAT)			2.268				
Aesthetics (AES)							0.017

Table 9
Construct the prediction summary.

Construct	$Q^2_{predict}$	RMSE	MAE
LOY	0.706	0.549	0.409
SAT	0.738	0.520	0.360

predictive performance, as lower values indicate more minor prediction errors. These results demonstrate that the model possesses strong predictive relevance and accuracy for the Loyalty and Satisfaction constructs, underscoring its effectiveness in explaining and predicting these critical outcomes in the study context. [Table 9](#) presents the construct prediction summary.

5. Discussion and implications

5.1. Discussion

This study offers significant insights into how AI-enabled service attributes influence customer satisfaction and loyalty among chain hotels in Thailand’s coastal tourism destinations. The findings reveal several key relationships that reshape our understanding of AI-enabled services in hospitality while challenging some established assumptions in the field. The analysis revealed three significant relationships that form the cornerstone of our findings. First, service efficiency positively affected customer satisfaction ($\beta = 0.251$, $p < 0.01$), indicating that efficient AI service delivery is crucial for guest satisfaction. This relationship highlights the importance of streamlined, effective AI interactions in hospitality. Second, enjoyment emerged as the strongest predictor of satisfaction ($\beta = 0.526$, $p < 0.001$), emphasizing the critical role of creating pleasant AI interactions. This finding underscores the importance of the hedonic aspects of AI services in hospitality settings. Third, the results indicated an exceptional relationship between satisfaction and customer loyalty ($\beta = 0.833$, $p < 0.001$), suggesting that satisfied customers will become loyal patrons.

Interestingly, and contrary to several previous studies, reliability, perceived intelligence, and aesthetics did not significantly influence customer satisfaction. This finding presents a notable departure from the existing literature. [Belanche et al. \(2021\)](#) and [Hsu and Lin \(2023\)](#) found that these factors were significant in other hospitality contexts. Our model also had strong explanatory power (R^2 values of 0.765 for satisfaction and 0.694 for loyalty), suggesting that focusing on efficiency and enjoyment was more crucial than previously thought in driving customer satisfaction and loyalty in AI-enabled hotel services.

The nonsignificant relationships between reliability, perceived intelligence, aesthetics, and satisfaction warrant careful examination because these findings challenge established theoretical frameworks. Previous research results positioned these factors as crucial determinants of service quality and satisfaction ([Belanche et al., 2021](#); [Parasuraman et al., 1988](#)) but our empirical analysis revealed no significant relationships (reliability: $\beta = -0.039$, $p = 0.696$; perceived intelligence: $\beta = 0.128$, $p = 0.108$; aesthetics: $\beta = 0.107$, $p = 0.168$).

This deviation from theoretical predictions suggests evolving customer expectations in AI-enabled services. Recent studies indicate that reliability has become a baseline requirement rather than a differentiator in service quality (Choi et al., 2024; Liu et al., 2024). Similarly, the nonsignificant effect of perceived intelligence challenges assumptions about the importance of AI sophistication (Chen et al., 2023), suggesting that customers prioritize functional benefits over advanced capabilities. The lack of impact from aesthetics aligns with Wong et al.'s (2024) findings that visual appeal may be secondary to performance in driving satisfaction.

These results have significant implications for theory development and resource allocation. First, established service quality frameworks may require refinement when applied to AI-enabled hospitality services, particularly in emerging markets and coastal tourism contexts. Second, hotels may need to reconsider investing in advanced AI capabilities or sophisticated interfaces to enhance customer satisfaction. Focusing resources on improving service efficiency and creating enjoyable interactions will yield better returns (Al-Hyari et al., 2023; Meeprom et al., 2024), especially in contexts where operational constraints and cultural factors influence AI service evaluation (Hansasooksin et al., 2024).

Our findings also suggest potential risks in overemphasizing technical sophistication at the expense of basic functionality. While organizations might be tempted to invest in cutting-edge AI features, such investments may not translate into increased customer satisfaction or loyalty without enhancing service efficiency or user enjoyment (Chen et al., 2023; Pizzi et al., 2023).

Both alignments and contrasts emerge when comparing our results with recent studies. The strong influence of service efficiency aligns with Chen et al.'s (2023) finding that AI-chatbot efficiency positively affects hotel customer satisfaction. Similarly, the importance of enjoyment supports Meeprom and Suttikun's (2024) research, highlighting the role of positive engagement in AI-based services, while the nonsignificant effect of reliability contrasts with Liu et al. (2024) who found that reliability was crucial for AI-enabled hotel services. This divergence could be attributed to the unique characteristics of coastal tourism destinations or cultural differences in the Thai market. Our findings regarding the strong relationship between satisfaction and loyalty align with recent studies by Al-Hyari et al. (2023) and Wong et al. (2024) who found robust connections between these constructs in luxury hotels; however, our study extends this understanding by demonstrating that this relationship holds specifically in the context of AI-enabled services in coastal tourism destinations. The nonsignificant impact of perceived intelligence contrasts with Holendova et al. (2024), who emphasized the importance of AI intelligence in shaping customer perceptions. Varying customer expectations in different market segments or cultural contexts might explain this difference. Similarly, the lack of significance for aesthetics differs from Pizzi et al.'s (2023) research, which found that visual appeal was crucial in AI interactions.

This study makes several novel contributions to the existing knowledge. First, it demonstrates that the functional and hedonic aspects (efficiency and enjoyment) of AI services in coastal tourism destinations may be more critical than technical sophistication or aesthetic appeal. This finding challenges the conventional wisdom that perceived intelligence and visual design are crucial for AI service success, as suggested by previous studies (Al-Hyari et al., 2023; Wong et al., 2024). Second, the strong mediating role of satisfaction in the relationship between service attributes and loyalty ($\beta = 0.438$ for enjoyment and $\beta = 0.209$ for efficiency) provides new insights into how AI services influence customer loyalty. This finding extends the work of Jeon (2024) and Niu and Mvondo (2023) by highlighting the pathways through which AI service attributes translate into loyalty outcomes. Third, we provide empirical evidence for the varying importance of different AI service attributes in a specific geographic and market context. Our findings suggest that the effectiveness of AI-enabled services may be more context-dependent than previously thought, supporting the need for localized approaches to AI implementation in hospitality.

This study addresses the research gap regarding the lack of understanding of how AI-enabled services affect specific geographic and market contexts, particularly in emerging markets like Thailand. Our results suggest that the effectiveness of AI-enabled services may vary significantly based on local market conditions and cultural contexts. This finding supports the need for more nuanced, context-specific approaches to AI implementation in hospitality, and also contributes to filling the gap in understanding how different AI service attributes influence customer loyalty in chain hotels within coastal tourism destinations. This study demonstrates the importance of efficiency and enjoyment over technical sophistication and aesthetics, providing valuable insights for hotel managers in similar contexts.

5.2. Theoretical implications

From a theoretical standpoint, the research contributes to the expanding literature on AI in hospitality by integrating multiple theoretical frameworks, including the TAM, SERVQUAL, S-O-R, and HVL models. This integrated approach provides a more comprehensive understanding of how AI-enabled service attributes influence customer satisfaction and loyalty. The findings partially support and extend existing theories while challenging some established concepts. The strong positive correlation between service efficiency and satisfaction aligns with the TAM and SERVQUAL models. This result emphasizes the importance of perceived usefulness and service quality in technology adoption; however, the nonsignificant effects of reliability, perceived intelligence, and aesthetics on satisfaction challenge the elements of these models and the HVL framework. This outcome indicates that in the context of AI-enabled services in chain hotels, functional aspects (particularly efficiency) may be more critical than perceived intelligence or visual appeal, which contradicts some previous findings (Belanche et al., 2021; Hsu & Lin, 2023). The significant role of enjoyment in affecting satisfaction and its indirect effect on loyalty through satisfaction supports the S-O-R model. This finding highlights the importance of emotional responses in influencing customer behavior and extends the research of Hernandez-Ortega and Ferreira (2021) on smart voice assistants to the broader context of AI-enabled hotel services. The study also emphasizes satisfaction's crucial role in the relationship between service attributes—especially enjoyment and efficiency—and customer loyalty. This finding enhances our understanding of how AI-enabled services influence customer loyalty, building upon previous research on the satisfaction-loyalty link within the hospitality industry (Al-Hyari et al., 2023; Liu et al., 2024). Furthermore, this study advances the theoretical understanding of AI-enabled services within specific market segments, notably chain hotels in coastal tourism destinations. The findings suggest that the effectiveness and importance of various AI service attributes may differ based on the hospitality context, contributing to a more nuanced understanding of AI adoption across different hotel types and locations.

5.3. Practical implications

This research offers several important implications for hotel managers and marketers. The significant influence of service efficiency and enjoyment on customer satisfaction and loyalty suggests that hotels should prioritize these aspects when implementing AI-enabled services. Managers should focus on developing AI systems that provide quick and accurate responses while creating enjoyable interactions for guests. While reliability, perceived intelligence, and aesthetics should not be overlooked, their nonsignificant effects might not be the primary drivers of satisfaction and loyalty; thus, hotels may benefit from reallocating resources away from enhancing perceived intelligence or visual appeal toward improving efficiency and enjoyment. The solid mediating role of satisfaction in the relationship between service attributes and loyalty highlights the importance of continuously monitoring and improving guest satisfaction with AI-enabled services. Hotels should implement

robust feedback mechanisms and regularly assess guest perceptions of AI interactions to maintain high satisfaction levels and foster long-term loyalty. This study focused on chain hotels in coastal tourism destinations, providing specific insights for this market segment. Managers of such properties should consider the unique characteristics of their guests' demographics and travel purposes when implementing AI services, as the effectiveness of these technologies may differ from those in other hotel types or locations.

The findings also have implications for AI developers and technology providers in the hospitality industry. The emphasis on efficiency and enjoyment suggests that future AI systems should be designed with these factors, potentially incorporating gamification or personalization elements to enhance the user experience. Furthermore, the study emphasizes the need for a balanced approach to implementing AI services in hotels. AI can significantly enhance efficiency and create enjoyable interactions; however, the importance of the human touch in the hospitality industry should not be overlooked. Hotels should find the right balance between AI-enabled services and human interactions to maximize guest satisfaction and loyalty. Finally, this research provides valuable insights for policymakers and industry associations in the hospitality sector. The findings can help inform guidelines or best practices for AI implementation in hotels, particularly coastal tourism destinations. This approach could ensure that AI adoption in the industry aligns with guest preferences, contributing to overall satisfaction and loyalty.

6. Conclusion

This study offers valuable insights into the impact of AI-enabled services on customer satisfaction and loyalty among chain hotels in the coastal tourism destinations of Western Thailand. The key findings reveal that service efficiency ($\beta = 0.251$, $p < 0.01$) and enjoyment ($\beta = 0.526$, $p < 0.001$) significantly influence customer satisfaction with AI-based services. Moreover, customer satisfaction strongly affects customer loyalty ($\beta = 0.833$, $p < 0.001$). The model exhibits strong explanatory power, with R^2 values of 0.765 for satisfaction and 0.694 for loyalty; thus, it accounts for a substantial portion of the variance in these critical outcomes. Interestingly, reliability, perceived intelligence, and aesthetics did not significantly influence satisfaction in this context, challenging some existing assumptions in the field. This research contributes to the growing literature on AI in hospitality by incorporating various theoretical frameworks, including the TAM, SERVQUAL, S-O-R, and HVL models. This integrated approach provides a more comprehensive understanding of how AI-enabled service attributes impact customer satisfaction and loyalty, particularly among chain hotels in coastal tourism destinations. This study also fills a notable research gap by offering empirical evidence on the effectiveness of AI-enabled services in a specific geographic and market context, specifically emerging markets like Thailand. The findings emphasize the essential roles of service efficiency and enjoyment in driving customer satisfaction and loyalty, simultaneously revealing unexpected nonsignificant effects that challenge other assumptions in the field. These insights establish a foundation for future research and provide valuable guidance for hotel managers to leverage AI technologies to enhance customer experiences and foster long-term loyalty in an increasingly competitive market.

7. Limitations and future directions

This study offers valuable insights into the influence of AI-enabled services on customer satisfaction and loyalty in chain hotels; however, several limitations should be noted. First, the research focuses on a specific geographic area (coastal tourism destinations in Western Thailand), which may restrict the generalizability of its findings to other regions or types of hospitality settings. The unique aspects of coastal tourism and Thai culture could shape customer perceptions and behaviors in ways that might not apply universally. Second, the cross-

sectional nature of the data collection provides only a snapshot of customer attitudes at one point in time, which limits our ability to observe how perceptions of AI services may change or evolve with technological advancements. Third, the study relies on self-reported measures, which could be influenced by social desirability bias or the respondents' ability to recall and accurately evaluate their experiences with AI services.

Despite these limitations, the study has notable strengths. Integrating multiple theoretical frameworks (TAM, SERVQUAL, S-O-R, and HVL) offers a comprehensive perspective on AI-enabled services in hospitality. The robust sample size ($N = 186$) and rigorous statistical analysis, including SEM and mediation analysis, enhance the reliability of the findings. Additionally, focusing on chain hotels within a specific market segment provides depth and context that is often lacking in broader industry studies. Future research could address these limitations and expand upon the current findings in various ways. For example, longitudinal studies could track changes in customer perceptions and loyalty toward AI-enabled services over time, providing insights into the long-term effects of these technologies. Cross-cultural comparisons could explore how AI service attributes influence satisfaction and loyalty across different geographic and cultural contexts. Experimental designs might isolate the effects of specific AI features or interface designs on customer experiences, addressing the potential confounding factors present in field studies.

Furthermore, given the unexpected nonsignificant effects of reliability, perceived intelligence, and aesthetics, future research should investigate the reasons behind these findings. Qualitative studies could offer deeper insights into customer expectations and experiences with AI in hospitality settings. Additionally, future research could examine potential moderating factors, such as customer tech-savviness or trip purpose, that could influence the relationship between AI service attributes and satisfaction. As AI technology advances, future studies should investigate how more sophisticated AI applications, such as personalized recommendations or predictive services, affect customer experiences and loyalty. The ethical implications of AI use in hospitality, including privacy concerns and the balance between automation and human interaction, also merit further exploration. Finally, future research could include other stakeholders, such as hotel staff and management, to better understand AI integration in the hospitality industry. Studies examining AI implementation's operational and financial impacts would complement the current emphasis on customer perceptions and behaviors. By addressing these limitations and pursuing the proposed future directions, researchers can enhance our understanding of AI's role in shaping customer experiences and loyalty within the evolving landscape of hospitality services.

CRedit authorship contribution statement

Noptanit Chotisarn: Writing – original draft, Validation, Supervision, Resources, Investigation, Data curation, Conceptualization. **Thadathibesra Phuthong:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Declaration of competing interest

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